Supplementary Information

A Path Is Not a State: Exact Predictive Refinement of Fixed Connectome Partitions

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This document retains audit-domain, diagnostic, and finite-data details supporting the journal manuscript. It does not enlarge the theorem domain or replace the published evidence identities of RIME Papers XVI and XVII.

S1. Distinct Audit Domains

The static coverage audit uses a nonnegative full-release edge table. The same-carrier liftability and first-order comparison use the signed, normalized annotation-intersection carrier. The reached-set follow-up uses 711 selected unit sources and exact rational orbits through horizon four. These are different finite domains even when they refer to the same underlying connectome release.

Table S1. Audited domains, observations, and evidence used in the manuscript.

Audit	Domain and observation	Evidence
Static coverage	Three nested sector-universe policies on nonnegative adjacency	Exact finite route counts
Same-carrier liftability	Signed normalized carrier; somaSide and superclass; depths 2 and 3	Exact finite product decisions
First-order scan	Eligible unit-state domain; superclass or somaSide means	Published numerical successor differences
Exact first-order obstruction	Two previously recorded somaSide unit sources	Two-row rational certificate

Audit	Domain and observation	Evidence
Profile-heterogeneity census	All 163,439 eligible unit sources; both registered mean observations	Exact class, source-pair, and coordinate-separation counts
Named one-lag obstruction	Declared history-fiber search; superclass means	Exact named-sector witness
Reached-set follow-up	711 selected sources; superclass means; horizon 4	Exact rational retained-orbit audits

The signed liftability result is an exact finite computational certificate. The previously published first-order scan is numerical; the added two-row certificate proves one exact inequality for the rational model, not the exactness of every scan entry. The profile census groups rational profiles through their canonical integer keys and uses bounded exact integer accumulation. It covers eligible unit sources, not arbitrary later activity vectors. The reached-set follow-up starts from retained exact-orbit artifacts and does not regenerate microscopic trajectories from the upstream connectome.

S2. Diagnostic Sequence

The procedure fixes a carrier, its action convention, an update, an observation, source states, and a finite recorded horizon. Static route tests are separate from dynamic factorization tests.

Table S2. Finite diagnostic sequence for comparing observation histories.

Step	Check and retained output	Interpretation controlled
Fix the domain	Choose one common set of source-time addresses for every history order	A changing left time boundary is not a memory effect

Step	Check and retained output	Interpretation controlled
Compare history partitions	Compare fiber memberships and successor constancy, not only value counts	Equal partitions mean added lags do not distinguish more points on that domain
Inspect residual noninjectivity	Separate singleton fibers from distinct states sharing an observation; inspect the update	A passing fiber test may be sustained by common leakage after drive clipping
Check iteration	Test successor membership in the verified image and factor identity on successive microscopic domains	A one-step factor need not define an image endomap

For a negative first-order result, retain a source pair and an exact successor-coordinate difference where the arithmetic permits it. For a positive result, retain the tested domain and complete fiber membership. Neither is a certificate for untested sources or times.

S3. Coverage Increments

The nonnegative depth-two audit uses three nested sector-universe policies. Here `<missing>` denotes a body whose curated annotation value is absent; `<unannotated>` denotes a body absent from the curated annotation table. Typed core excludes the latter label at all three positions. Relay strict allows it at endpoints, and coverage inclusive also allows it at the middle body. These policy increments therefore admit different candidate populations.

Table S3. Incremental supported and lifted triples under policy enlargement.

Enlargement	Support gain	Lift gain
Typed core to relay strict	782	781
Relay strict to coverage inclusive	812	374

The first increment has one nonlifting candidate. In the second, 438 of the 812 new triples have no common microscopic relay. The lifted count increases in both increments, although the overall liftability fraction falls in the second. For old supported and lifted counts B, L and a new increment $\Delta B > 0, \Delta L$, the ratio difference is the elementary identity

$$\frac{L + \Delta L}{B + \Delta B} - \frac{L}{B} = \frac{\Delta B}{B + \Delta B} \left(\frac{\Delta L}{\Delta B} - \frac{L}{B} \right).$$

The audit checks incoming/outgoing relay intersections body by body. It does not infer a causal effect of missing annotation.

S4. Historical First-Reference Scan

Within each observation sector, the registered first-reference scan selects the first eligible node in the source order and compares later nodes with that reference. Its witness count is not an all-pairs count or a prevalence estimate; the defect is one witnessed discrepancy, not the diameter of a fiber.

Table S4. Published first-reference one-step observation defects.

Partition	Sectors	Witnesses	Largest L1 defect
superclass	28	141,296	0.005651574473947287
somaSide	4	163,435	0.0004874856094383272

The defect magnitudes do not rank the partitions, which differ in dimension, sector sizes, and compression.

S5. Coordinate-Resolved Pair Counts

For source sector E_i and target coordinate j, let n_{ijq} count sources with profile coordinate q. The exact number of unordered same-observation source pairs separated by j is

$$D_{ij} = \binom{N_i}{2} - \sum_q \binom{n_{ijq}}{2}.$$

Table S5. Absolute somaSide pair separations by target coordinate.

Target coordinate	Separated within-sector source pairs
L	5,008,763,423
R	4,993,360,210
`<missing>`	3,413,153,081
M	1,307,422,058

The common denominator is 5,432,178,301 unordered within-sector pairs. Pairs separated by several coordinates appear in several rows, so the counts are not additive. The manuscript's Figure 2 visualizes normalized percentages; these exact counts are its source-resolved counterpart.

S6. Exact Histories on the Common Reached Set

The Paper XVII follow-up selected 711 of 163,439 eligible sources across 25 nonempty source strata. One source per stratum was retained before a capacity-based largest-remainder allocation; SHA-256 ranking then selected members within strata. The selection rule did not inspect trajectory outcomes. Exact rational orbits were retained through horizon four.

The original history-order comparison used different left boundaries: current observations at $t = 0, \ldots, 3$, one lag at $t = 1, \ldots, 3$, and two lags at $t = 2, 3$. The common comparison domain is $D_0 \times \{2, 3\}$.

Table S6. History-fiber counts on the common 1,422-window domain.

History order	Windows	Distinct history values	Largest fiber	Failure fibers
0	1,422	1,228	51	0
1	1,422	1,228	51	0
2	1,422	1,228	51	0

The three partitions, not merely their summary counts, are equal on this domain. Six unsafe fibers at $t = 1$ contain 51 sources; all have singleton observations at $t = 2$ in the selected source set. Five separate persistent cohorts of sizes 51, 25, 19, 4, and 3 remain noninjective with the same membership at $t = 2$ and $t = 3$. These slices account for $2[(51 - 1) +$

$(25-1)+(19-1)+(4-1)+(3-1)] = 194$ redundant source-time windows. The manuscript's Appendix A gives the distinct-state and drive-off interpretation of these retained cohorts.

For the earlier superclass one-lag audit, equality was checked using the canonical integer key encoding the rational profile

$$(i(a), s_a/g_a, p_{a1}/g_a, \dots, p_{am}/g_a), \qquad g_a = \gcd(s_a, p_{a1}, \dots, p_{am}).$$

A named-sector follow-up found an exact witness in `cb_intrinsic`: the one-lag histories agree while nine next-observation coordinates differ. This is a separate domain from the 711-source positive follow-up.

S7. Reproduction and Provenance

The published source packages remain independently identified as Paper XVI and Paper XVII. Paper XVI binds upstream MaleCNS data, selected carrier files, static audits, and dynamic records. Paper XVII starts its public checks from retained exact-orbit artifacts and transition-cache records.

The predictive-profile Dataset (DOI: 10.5281/zenodo.23555888) contains the exact full-domain census, its source and seven focused controls, the pinned derived carrier, and both figure sources. Its default verifier checks the full file inventory and byte digests; optional source replay recomputes the census. The separate two-source witness supplement contains the exact certificate, its checker, six controls, and the three Paper XVI metadata records it reads. Its default check verifies compact arithmetic without the large carrier. To re-extract the two source rows, pass the carrier directory from the extracted Dataset to the witness verifier's `--replay --carrier-dir` option. A compact check alone does not establish that the rows were extracted from the carrier. Neither replay is independent replication or reconstruction of the original upstream connectome pipeline.